

Experiment 4 – Qualys Whitelisting and Connectivity Validation in Kali Linux

Aim

To configure outbound firewall whitelisting for Qualys communication from Kali Linux and validate DNS, HTTPS and TCP connectivity.

Procedure

- 1. Check Kali network configuration and Internet/DNS connectivity.
- 2. Check/install UFW.
- 3. Deny incoming traffic, allow outgoing traffic and enable UFW.
- 4. Add the Qualys IP, CIDR and listed Cloudflare HTTPS whitelist rules.
- 5. Verify numbered UFW rules. • 6. Test Qualys DNS resolution.
- 7. Test HTTPS connectivity with curl.
- 8. Test TCP 443 with nc.
- 9. Remove one rule using its actual rule number and test connectivity.
- 10. Restore the rule, retest, enable UFW logging and inspect logs.

Commands

```
ip addr
ip route
ping -c 4 8.8.8.8
nslookup google.com
sudo ufw status
sudo apt update
sudo apt install ufw -y
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw enable
sudo ufw status verbose
sudo ufw allow out to 168.138.113.116 port 443
proto tcp
sudo ufw allow out to 150.230.234.34 port 443
proto tcp
sudo ufw allow out to 103.216.98.0/24 port 443
proto tcp
```

```
sudo ufw allow out to 103.75.173.0/24 port 443
proto tcp
sudo ufw allow out to 162.159.152.21 port 443
proto tcp
sudo ufw allow out to 162.159.153.243 port 443
proto tcp
sudo ufw status numbered
nslookup qualysguard.qg1.apps.qualys.in
nslookup distribution.qg1.apps.qualys.in
curl -I https://qualysguard.qg1.apps.qualys.in
curl -I https://distribution.qg1.apps.qualys.in
nc -vz qualysguard.qg1.apps.qualys.in 443
sudo ufw delete <rule-number>
sudo ufw logging medium
sudo journalctl -k | grep UFW
```

```
➔ cd whitelist
(mohana@kali)~[/Documents/Blockchain/whitelist]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a8:40:95 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 59201sec preferred_lft 59201sec
    inet6 fd17:625c:f037:2:f80e:4ded:6918:e35e/64 scope global temporary dynamic
        valid_lft 86311sec preferred_lft 14311sec
    inet6 fd17:625c:f037:2:a00:27ff:fea8:4095/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86311sec preferred_lft 14311sec
    inet6 fe80::a00:27ff:fea8:4095/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether fa:5c:8c:4e:0f:28 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever

(mohana@kali)~[/Documents/Blockchain/whitelist]
$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown

(mohana@kali)~[/Documents/Blockchain/whitelist]
$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.

 8.8.8.8 ping statistics:
 4 packets transmitted, 0 received, 100% packet loss, time 3081ms

(mohana@kali)~[/Documents/Blockchain/whitelist]
$ nslookup google.com
Server:      10.0.2.3
Address:     10.0.2.3#53
```

```
(mohana@kali)-[~/Documents/Blockchain/whitelist]
$ nslookup google.com
Server:      10.0.2.3
Address:     10.0.2.3#53

Non-authoritative answer:
Name:   google.com
Address: 192.178.173.113
Name:   google.com
Address: 192.178.173.139
Name:   google.com
Address: 192.178.173.102
Name:   google.com
Address: 192.178.173.101
Name:   google.com
Address: 192.178.173.138
Name:   google.com
Address: 192.178.173.100
Name:   google.com
Address: 2404:6800:4007:821::200e

(mohana@kali)-[~/Documents/Blockchain/whitelist]
$ sudo ufw default deny incoming
[sudo] password for mohana:
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)

(mohana@kali)-[~/Documents/Blockchain/whitelist]
$ sudo ufw default deny outgoing
Default outgoing policy changed to 'deny'
(be sure to update your rules accordingly)

(mohana@kali)-[~/Documents/Blockchain/whitelist]
$ sudo ufw enable
Firewall is active and enabled on system startup

(mohana@kali)-[~/Documents/Blockchain/whitelist]
$ sudo ufw allow out to 168.138.113.116 port 443 proto tcp
Skipping adding existing rule
```

```
(mohana@kali)-[~/Documents/Blockchain/whitelist]
$ sudo ufw allow out to 103.216.98.0/24 port 443 proto tcp
Skipping adding existing rule

(mohana@kali)-[~/Documents/Blockchain/whitelist]
$ sudo ufw allow out to 162.159.152.21 port 443 proto tcp
Skipping adding existing rule

(mohana@kali)-[~/Documents/Blockchain/whitelist]
$ sudo ufw allow out to 162.159.153.243 port 443 proto tcp
Skipping adding existing rule

(mohana@kali)-[~/Documents/Blockchain/whitelist]
$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), deny (outgoing), deny (routed)
New profiles: skip
```

To	Action	From
--		
53/udp	ALLOW OUT	Anywhere
53/tcp	ALLOW OUT	Anywhere
168.138.113.116 443/tcp	ALLOW OUT	Anywhere
150.230.234.34 443/tcp	ALLOW OUT	Anywhere
103.216.98.0/24 443/tcp	ALLOW OUT	Anywhere
103.75.173.0/24 443/tcp	ALLOW OUT	Anywhere
162.159.153.243 443/tcp	ALLOW OUT	Anywhere
162.159.152.21 443/tcp	ALLOW OUT	Anywhere
53/udp (v6)	ALLOW OUT	Anywhere (v6)
53/tcp (v6)	ALLOW OUT	Anywhere (v6)

Result:

Thus configuring outbound firewall whitelisting for Qualys communication from Kali Linux and validate DNS, HTTPS and TCP connectivity was done successfully.